Anand Nagaraj

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Profile Summary

Master's student in Design and Test of Integrated Circuits with 3+ years of industry experience in digital functional verification, SystemVerilog/UVM testbenches, and AI-assisted EDA automation. Skilled in constructing modular UVM components (scoreboards, drivers, monitors), authoring SystemVerilog Assertions (SVA) for protocol and formal property checks, and deploying Python pipelines for log diagnostics and coverage closure. Eager to contribute hands-on verification and research in AI-driven verification methods as a Working Student in Digital Functional Verification at Infineon Dresden (Job ID: HRC1851190).

Education

Technische Universität Chemnitz **Apr 2026 – Present**
M.Sc. in Design and Test of Integrated Circuits Chemnitz, Germany
Core Modules: Verification of Digital Systems, Elements of Integrated Circuits, Design of Heterogeneous Systems, Digital Components and Architectures, Test and Design of Mixed-Signals.

Visvesvaraya Technological University (VTU / BMSIT) **Aug 2018 – Apr 2022**
B.E. in Electronics & Telecommunication Engineering Bengaluru, India
Core Curriculum: Electronic Devices & Circuits, CMOS VLSI Design, Computer Architecture, Digital Systems, Microprocessors.

Technical Skills

Verification Methodologies: UVM (Scoreboards, Checkers, Sequences, Drivers, Monitors), OOP Concepts, pyUVM, Cocotb, Constrained-Random Verification (CRV), Functional Coverage (Covergroups, Coverpoints)
HDLs, HVLs & Formal Assertions: SystemVerilog (IEEE 1800), Verilog, VHDL, SystemVerilog Assertions (SVA), Linting, Formal Property Verification Concepts
EDA Tools & Simulators: Cadence Xcelium, Synopsys VCS, Siemens QuestaSim, ModelSim, Synopsys Design Compiler, GTKWave
AI-Assisted Verification & Automation: Python Pipelines for EDA Simulation Log Parsing, LLM-Driven Script Generation, Automated Warning/Error Clustering, Coverage Metric Extraction
IC Fundamentals & Architectures: Cross-Clock Domain (CDC) Validation, SoC Interconnects, RISC-V (RV32I/M), AMBA (AHB/APB), PCIe TLP, Testmode Architectures
Scripting & Environments: Python (OOP, Regression Automation), Bash/Shell Scripting, Linux/Unix, Git
Languages: English (Fluent, C1 Technical Documentation & Reporting), German (Conversational, A2)

Professional Work Experience

IndusIP Labs Pvt Ltd Bengaluru, India
VLSI Verification & Design Engineer (Founding Team) **Nov 2024 – Feb 2026**

- Constructed simulation verification environments using SystemVerilog UVM and Python Cocotb to validate multi-port Network-on-Chip (NoC) switches under heavy synthetic traffic patterns.
- Developed UVM scoreboards, transaction checkers, and monitors to detect functional discrepancies, packet dropouts, and routing latency violations.
- Authored SystemVerilog Assertions (SVA) to monitor cross-clock domain (CDC) signals, verifying deadlock-free arbitration and protocol compliance across asynchronous FIFO boundaries.
- Built scalable regression test suites using Python scripts, automating test execution, aggregating simulation transcripts, and extracting functional coverage metrics.
- Leveraged AI-assisted workflows and scripting to accelerate testbench boilerplate generation and cluster simulation error logs for rapid root-cause analysis.

Sinergia Media Labs (Simelabs) Kochi / Bengaluru, India
RTL Design & Verification Engineer **Sep 2022 – Oct 2024**

- Developed modular UVM verification components (agents, drivers, monitors, scoreboards) for high-speed PCIe Transaction Layer IP, validating packet formatting, TLP ordering, and buffer management.
- Implemented custom scoreboard checkers to validate credit-tracking flow control, injecting corner cases such as credit starvation and back-to-back burst transactions.
- Built Python Cocotb co-simulation testbenches to verify register configuration read/write operations and validate hardware-software interface boundaries.
- Spearheaded an AI-assisted automation initiative (ChipGPT), building Python agent pipelines to parse EDA simulation logs, cluster warning patterns, and streamline regression triage.
- Authored structured verification plans, testmode checklists, and technical verification reports to ensure transparency across multidisciplinary engineering teams.

Key Projects

Moola-V (MR5): 5-Stage Pipelined RV32IM Core Verification Active Open-Source Project
Architectural Compliance, Cocotb Self-Checking Testbench & SVA GitHub: moola_riscv_core

- Architected a synthesizable 5-stage in-order RV32IM processor core in SystemVerilog, incorporating dedicated ALU, multiplier/divider units, and hazard detection logic.
- Built an automated Cocotb self-checking co-simulation test environment running lock-step against the golden Spike Instruction Set Simulator (ISS).
- Embedded SystemVerilog Assertions (SVA) and functional covergroups to check pipeline stage validity, hazard stall signals, and exception triggers.
- Attained 100% architectural compliance across all 49 RISCOF test suites (RV32I, RV32M, hints), maintaining a regression suite for rapid feature validation.

Mixed-Signal Frequency Measurement & Interface System Academic Project, TU Chemnitz
VHDL / VHDL-AMS Heterogeneous Architecture & Multi-Clock CDC Design

- Architected a reliable dual-clock digital measurement unit in VHDL, implementing cooperating FSMs across asynchronous domains with double-flop synchronizers and request-acknowledge CDC handshake protocols.
- Modeled a continuous-time MEMS analog front-end (sensor, RC low-pass filter, and VCO) in VHDL-AMS, integrating it with digital threshold decision logic and a 115200-baud UART transmitter for end-to-end verification.

Parameterized Dual-Port SRAM Macro Subsystem GitHub: tt_moola_dp_sram
Memory Architecture, Transistor-Level Characterization & Verification

- Designed a 16×8 parameterized dual-port synchronous SRAM macro module on the SkyWater 130nm process using the Tiny Tapeout open-source ASIC flow.
- Analyzed 6T bit-cell transistor sizing (W/L ratios) and characterized Read/Write Static Noise Margins (SNM) in SPICE across supply voltage variations.
- Verified memory write-masking, read latency, and simultaneous access cycles using SystemVerilog testbenches prior to GDSII creation.

Honors & Key Achievements

- **Deutschlandstipendium Scholarship Recipient (2026 – 2027):** Awarded by TU Chemnitz and sponsored by Infineon Technologies Dresden for academic excellence and outstanding potential.
- **SEMICON Europa × TUM Venture Labs Hackathon (NextSEMI 2026):** Selected as one of only 25 participants across Europe for an intensive semiconductor prototyping and hardware sprint in Munich (Nov 2026).
- **State Merit Scholarship (Karnataka Board of Education):** Awarded 100% undergraduate tuition waiver for ranking among the top candidates in the state competitive examination.
- **Best Outgoing Student Award (2022):** Conferred by the Department of Electronics & Telecommunication Engineering, VTU/BMSIT, for sustained academic and leadership excellence.
- **District Kabaddi Championship:** Secured 2nd place in a competitive Under-19 District Tournament.

Publications

Peripheral Shared Memory Over PCI Express Link **Oct 2025**
Presented at the 7th International Conference on Innovative Data Communication Technologies and Application (ICIDCA); analyzes shared memory throughput, buffer sizing trade-offs, and transaction latency across high-speed PCIe links.

Professional Training & Interests

Advanced VLSI Design and Verification: 6-month full-time training at Maven Silicon (SystemVerilog, UVM, Verilog RTL, AMBA Protocols, QuestaSim/ModelSim Flow).
Interests: Distance Running, Cricket, Table Tennis, Strength Training.